Department of Information and Communication Technology
Islamic University, Kushtia, Bangladesh

PRACTICAL EXPERIMENT ASSIGNMENT

Course Title	Mobile and Cellular Communication Laboratory
Course Code	ICT-4204
Practical Assignment	Three experiments are assigned to each student for practice. One experiment from the assigned set will be selected for the final practical examination.

Assignment Policy

The experiment allocation has been generated using a predefined systematic sequence and rotation logic. The assignments were not made manually by considering any individual student, roll number, academic performance, or personal preference. A computer-assisted arrangement, with assistance from ChatGPT, was used to distribute the experiments in a structured and balanced manner. Therefore, no student has intentionally been assigned an easier or more difficult experiment.

Allocation Logic

The allocation follows an ascending cyclic pattern across groups: Group A starts from Experiment 1, Group B from Experiment 2, Group C from Experiment 3, Group D from Experiment 4, Group E from Experiment 5, and Group F from Experiment 6. After Experiment 9, the sequence continues again from Experiment 1. Each student's final practical experiment is selected from within the three experiments assigned to that student.

Experiment Code

Code	Experiment
E1	Calculation of Doppler Shift and Maximum Mobile Velocity
E2	Calculation of Co-channel Reuse Distance and Reuse Ratio
E3	Study of Frequency Allocation and Channel Assignment in Cellular Networks
E4	Study and Simulation of GSM Network Architecture
E5	Simulation of Handover Between Two Cellular Cells
E6	Received Signal Strength-Based Base Station Selection
E7	Free-Space Path Loss and Received Power Analysis Using MATLAB
E8	Analysis of Cell Coverage, Cell Radius, and Cell Splitting
E9	Cellular Traffic, Channel Capacity, and Erlang-B Blocking Probability Analysis

Group A

Roll	Name	Assigned 3 Experiments
2018001	ZUBAIR	E1 Doppler; E2 Reuse Ratio; E5 Handover
2018003	MAHABUB	E2 Reuse Ratio; E3 Frequency Allocation; E6 BS Selection
2018004	TONMOY	E3 Frequency Allocation; E4 GSM; E7 FSPL
2018005	JULFIKAR	E4 GSM; E5 Handover; E8 Cell Coverage
2018006	SHAWKAT	E5 Handover; E6 BS Selection; E9 Erlang-B
2018007	MEHEDI	E6 BS Selection; E7 FSPL; E1 Doppler
2018008	SAMRAT	E7 FSPL; E8 Cell Coverage; E2 Reuse Ratio
2018011	ADNANUL	E8 Cell Coverage; E9 Erlang-B; E3 Frequency Allocation
2018012	FAIZUL	E9 Erlang-B; E1 Doppler; E4 GSM
2018013	IMTIAZ	E1 Doppler; E5 Handover; E8 Cell Coverage

Group B

Roll	Name	Assigned 3 Experiments
2018014	SHEMANTO	E2 Reuse Ratio; E3 Frequency Allocation; E6 BS Selection
2018015	RONY	E3 Frequency Allocation; E4 GSM; E7 FSPL
2018016	SAIDUR	E4 GSM; E5 Handover; E8 Cell Coverage
2018017	TAREK	E5 Handover; E6 BS Selection; E9 Erlang-B
2018018	ABIR	E6 BS Selection; E7 FSPL; E1 Doppler
2018020	AZIZ	E7 FSPL; E8 Cell Coverage; E2 Reuse Ratio
2018021	TANVIR	E8 Cell Coverage; E9 Erlang-B; E3 Frequency Allocation
2018022	TARIF	E9 Erlang-B; E1 Doppler; E4 GSM
2018023	MILAD	E1 Doppler; E2 Reuse Ratio; E5 Handover
2018024	ARNOB	E2 Reuse Ratio; E5 Handover; E8 Cell Coverage

Group C

Roll	Name	Assigned 3 Experiments
2018025	ASHIKUR	E3 Frequency Allocation; E4 GSM; E7 FSPL
2018026	SHAHJAHAN	E4 GSM; E5 Handover; E8 Cell Coverage
2018027	SHIHABUL	E5 Handover; E6 BS Selection; E9 Erlang-B
2018028	RUHAN	E6 BS Selection; E7 FSPL; E1 Doppler
2018029	SHAHIN	E7 FSPL; E8 Cell Coverage; E2 Reuse Ratio
2018030	SOMPOD	E8 Cell Coverage; E9 Erlang-B; E3 Frequency Allocation
2018031	RONI	E9 Erlang-B; E1 Doppler; E4 GSM
2018032	HIMEL	E1 Doppler; E2 Reuse Ratio; E5 Handover
2018033	TITAS	E2 Reuse Ratio; E3 Frequency Allocation; E6 BS Selection
2018034	SAKIB	E3 Frequency Allocation; E6 BS Selection; E8 Cell Coverage

Group D

Roll	Name	Assigned 3 Experiments
2018035	SAJID	E4 GSM; E5 Handover; E8 Cell Coverage
2018036	SHAHINUR	E5 Handover; E6 BS Selection; E9 Erlang-B
2018037	SAZIDUR	E6 BS Selection; E7 FSPL; E1 Doppler
2018038	ULLASH	E7 FSPL; E8 Cell Coverage; E2 Reuse Ratio
2018040	SAJJAD	E8 Cell Coverage; E9 Erlang-B; E3 Frequency Allocation
2018041	HALIMA	E9 Erlang-B; E1 Doppler; E4 GSM
2018043	BITHI	E1 Doppler; E2 Reuse Ratio; E5 Handover
2018044	INDRANI	E2 Reuse Ratio; E3 Frequency Allocation; E6 BS Selection
2018045	SHUROVI	E3 Frequency Allocation; E4 GSM; E7 FSPL
2018047	FARZANA	E4 GSM; E6 BS Selection; E9 Erlang-B

Group E

Roll	Name	Assigned 3 Experiments
2018048	TANJU	E5 Handover; E6 BS Selection; E9 Erlang-B
2018049	ROMA	E6 BS Selection; E7 FSPL; E1 Doppler
2018050	SADIA	E7 FSPL; E8 Cell Coverage; E2 Reuse Ratio
2018051	MASUMA	E8 Cell Coverage; E9 Erlang-B; E3 Frequency Allocation
2018052	SHARMISHTHA	E9 Erlang-B; E1 Doppler; E4 GSM
2018053	SHITHY	E1 Doppler; E2 Reuse Ratio; E5 Handover
2018054	TAEEFA	E2 Reuse Ratio; E3 Frequency Allocation; E6 BS Selection
2018055	SRABONI	E3 Frequency Allocation; E4 GSM; E7 FSPL
2018056	KUMU	E4 GSM; E5 Handover; E8 Cell Coverage
2018057	RAISHA	E5 Handover; E7 FSPL; E9 Erlang-B

Group F

Roll	Name	Assigned 3 Experiments
1918005	SHAIDUR	E6 BS Selection; E7 FSPL; E1 Doppler
1918009	IBRAHIM	E7 FSPL; E8 Cell Coverage; E2 Reuse Ratio
1918011	ISMAEL	E8 Cell Coverage; E9 Erlang-B; E3 Frequency Allocation
1918019	SAYED	E9 Erlang-B; E1 Doppler; E4 GSM
1918020	RAHIQUL	E1 Doppler; E2 Reuse Ratio; E5 Handover
1918029	RAKIB	E2 Reuse Ratio; E3 Frequency Allocation; E6 BS Selection
1918055	NAIMUR	E3 Frequency Allocation; E4 GSM; E7 FSPL
1918057	SHADMAN	E4 GSM; E5 Handover; E8 Cell Coverage

Note: *Each student will perform one final practical experiment selected from the three experiments assigned to him/her. The selection may be made by the examination board or through a fair random draw during the examination.*